

Kshitij Lingthep (Chris L)

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EDUCATION | The University of Mississippi | Oxford, MS | Aug 2016 – May 2020

- **B.S. In Computer Science [Full ride Scholarship]** | Overall GPA: 3.71 | Honor's Roll | Provost Scholar

RESEARCH INTERESTS

- Machine Learning, Deep Learning, Natural Language Processing, Multimodal Learning, Generative AI

RESEARCH & AI PROJECTS

Recursive Fine-Tuning and Model Degradation in Biomedical QA systems | Research | August 2026 – Present | *Manuscript in preparation for conference submission*

- Investigated model degradation from **recursive fine-tuning** on synthetic biomedical QA data using **deep learning**, **LLMs**, **QLoRA**, and **Hugging Face Transformers** across multiple training generations and random seeds.
- Evaluated **lexical**, **semantic**, and **biomedical entity-level** changes using **PubMedQA**, **BERT** based semantic metrics, **biomedical NER**, and **statistical analysis**.
- Analyzed changes in **model diversity**, **domain knowledge retention**, and answer quality across recursive generations to characterize the effects of synthetic data accumulation.

Image Captioning and Model Evaluation | Independent Research | May 2026 - August 2026

- Developed a deep learning image captioning pipeline using **CNN** based image feature extraction and **LSTM** based sequence generation and built interactive interface using **Gradio UI**.
- Evaluated generated captions using **BLEU-1** through **BLEU-4** to measure caption quality and model performance.

Multi Agent Deal Discovery platform | Dec 2025 - Jan 2025

- Built an autonomous **multi-agent GenAI** system using **OpenAI** and **Perplexity** APIs, coordinating agents for deal **discovery**, **pricing**, **retrieval**, **prediction**, and **planning**.
- Developed production AI pipelines combining a fine-tuned LLM, RAG with **ChromaDB**, and a PyTorch model, delivered via a **Gradio UI** with structured outputs and real-time monitoring.

Find Price | Dec 2025 - Dec 2025

- Designed and delivered a GenAI system using **OpenAI** and **Perplexity** APIs, encompassing **data curation**, **preprocessing**, **prompt engineering**, **embeddings**, and **inference** to generate automated, cost-optimized product insights.
- Built **NLP pipelines** to extract and **semantically analyze** Amazon product descriptions, estimate prices, and recommend cheaper alternative products using embedding-based search and AI-driven reasoning.

ACADEMIC PROJECTS

Windows Portfolio | Jan 2023 - July 2023

- Developed a **Windows 11** themed [portfolio](#) website using **React** and **Tailwind**, with linting/code standards (ESLint/Prettier) and **CI/CD** via **GitHub Actions**.
- Implemented a dynamic [project gallery](#) with interactive modals and integrated **Google Analytics** to monitor user engagement and traffic.

Attendance Tracking Application | University of Mississippi | Jan 2020 - May 2020

- Developed a real time **React** based attendance tracking app for Department of Mathematics using **Firestore** for storage and authentication with role-based access control.
- Built a backend with **Node.js**, handling data processing, business logic, and **RESTful API** interactions with the frontend.

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, C#
- **Machine Learning & AI:** PyTorch, TensorFlow/Keras, Hugging Face Transformers, NLP, OpenAI API, Perplexity API, RAG, embeddings, ChromaDB, LLM, prompt engineering, NLP, semantic search
- **Deep Learning & Research:** QLoRA, LoRA, PEFT, 4-bit Quantization, BERT, Biomedical NER, BERTScore, BLEU, ROUGE, METEOR, Statistical Analysis
- **Software Development:** ReactJS, VueJS, Node.js, Spring Boot, .NET, REST API, Microservices, SignalR
- **Databases:** MySQL, PostgreSQL, MongoDB, Oracle, PL/SQL
- **Cloud & Development:** AWS, Docker, Kubernetes, Terraform, Git, GitHub Actions, GitLab CI/CD, Grafana

CERTIFICATIONS

- **AWS Certified Solutions Architect** – AWS, Terraform – LinkedIn, Generative AI – Udemy, MERN Stack – LinkedIn, React – TestDome, Project Management – Microsoft

Software Engineer | SK Battery America | May 2024 - Present, Stanton, TN

- Designed and evaluated real time software systems for battery manufacturing and facility monitoring to improve operational visibility, system reliability, and data driven decision support.
- Developed real-time monitoring dashboards using **React** and **SignalR** to visualize equipment status, production conditions, and operational alerts across manufacturing environments.
- Contributed to the design and implementation of **Manufacturing Execution System (MES)** for tracking production workflows, equipment status, and quality parameters.
- Led development and performance optimization for **Facility Monitoring System (FMS)** applications using **Spring Boot**, **ASP.NET MVC**, and **Oracle**, supporting highly available production systems with 99.9% uptime.
- Analyzed equipment logs and downtime patterns to identify recurring failure conditions and operational bottlenecks, contributing to a 15–20% reduction in unplanned production stoppages.
- Investigated backend performance and data processing bottlenecks by optimizing **PL/SQL** stored procedures and **Oracle** queries, improving reporting efficiency and application responsiveness.
- Developed monitoring and administrative tools for device management, logging, fault analysis, and production support, enabling faster diagnosis of equipment and application issues.
- Collaborated with engineering and manufacturing teams to translate operational problems into measurable system improvements and validated solutions in production environments.

R&D Software Engineer | Foxconn Industrial Internet | July 2020 - May 2024, Mount Pleasant, WI

- Worked on R&D projects involving smart manufacturing, facility monitoring, IoT, video streaming, temperature recognition, and EV charging management systems.
- Worked on real time video streaming and temperature recognition system during COVID 19 to support automated screening and facility monitoring.
- Developed interactive dashboards and data visualization interfaces using **React**, **Vue**, **Chart.js**, and **ECharts** to display real-time and historical manufacturing data.
- Improved frontend performance bottlenecks and applied techniques such as **lazy loading**, **code splitting**, **memoization**, and optimized state management, reducing page load times.
- Designed and analyzed microservice based architectures using **Spring Boot** and **.NET Core**, evaluating scalability, reliability, maintainability, and deployment performance across distributed systems.
- Built event-driven data pipelines using **Kafka** to support asynchronous communication and real-time processing across manufacturing applications.
- Developed and optimized data-access layers using **Hibernate**, **Spring Data JPA**, **Oracle**, and **MySQL**, with emphasis on query performance, data consistency, and system scalability.
- Conducted performance analysis and system monitoring using application logs, operational metrics, and infrastructure data to identify bottlenecks, recurring failures, and opportunities for system improvement.
- Contributed to cloud migration and architecture evaluation using **AWS** services such as **EC2**, **RDS**, **S3** etc.
- Developed reusable internal software libraries and shared components, reducing duplicated development effort and improving consistency across projects.
- Implemented automated testing and **CI/CD** pipelines using **JUnit**, **Mockito**, **Jest**, and **GitLab CI** to support reproducible testing, validation, and reliable software delivery.
- Analyzed user interaction and engagement data using **Google Analytics** to understand application usage patterns and inform interface and feature improvements.

Supplemental Instruction Leader | University of Mississippi | Jan 2019 - May 2020, Oxford, MS

- Led supplemental instruction sessions for advanced mathematics courses, reinforcing complex concepts through collaborative problem solving process.
- Developed practice problems and guided students through mathematical reasoning, problem solving strategies, and exam preparation.

Mathematics Tutor | University of Mississippi | Jan 2018 - May 2020, Oxford, MS

- Provided individual and small group tutoring in mathematics, helping students strengthen understanding of concepts and problem solving skills.
- Explained challenging mathematical concepts using step by step approaches tailored to students individual learning needs.